HUMAN HEALTH, IMMUNOLOGY & EVOLUTIONARY BIOLOGY

Comprehensive High-Yield Master Notes: NMC Rationalized Core Modules

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1 Principles of Immunology & Host Defense Mechanisms

Innate Immunity: Non-Specific First & Second Lines of Defense

Present from birth, non-specific, lacks immunological memory. Comprises four distinct anatomical and physiological barrier tiers:

1. **Physical Barriers:** Skin (keratinized stratum corneum prevents microbial ingress); Mucous coating of epithelial lining across respiratory, gastrointestinal, and urogenital tracts traps entrapped microbes.
2. **Physiological Barriers:** Gastric hydrochloric acid (pH $\approx 1.5 - 2.0$ destroys ingested bacteria); Saliva and Tears contain **lysozyme** (hydrolyzes $\beta(1 \to 4)$ glycosidic bonds in bacterial peptidoglycan cell walls).
3. **Cellular Barriers:** Phagocytes that engulf and digest foreign pathogens:
   - **PMNL** (Polymorphonuclear Leukocytes / Neutrophils): Most abundant circulating WBCs ($60-65\%$).
   - **Monocytes** ($6-8\%$) differentiating into tissue **Macrophages** (e.g., Kupffer cells, microglia, alveolar macrophages).
   - **Natural Killer (NK) Cells:** Specialized cytotoxic lymphocytes that destroy virus-infected and malignant tumor cells non-specifically via perforins and granzymes.
4. **Cytokine Barriers:** Virus-infected cells secrete non-viral glycoproteins called **Interferons** (α, β, γ), which bind to uninfected neighboring cells, inducing translation of antiviral proteins (AVPs) that inhibit viral translation and replication.

Acquired (Adaptive) Immunity: Humoral vs Cell-Mediated Immunity (CMI)

Characterized by antigenic specificity, diversity, discrimination between self and non-self, and immunological memory.

Property	Humoral / Antibody-Mediated Immunity (AMI)	Cell-Mediated Immunity (CMI)
Effector Cells	**B-lymphocytes** differentiating into Plasma cells (antibody factories) and Memory B cells.	**T-lymphocytes** maturing into Helper T ($CD4^+T_H$), Cytotoxic T ($CD8^+T_C$), and Regulatory T cells.
Mode of Action	Secretes soluble **Antibodies** (H_2L_2) into blood plasma, lymph, and extracellular fluids.	Direct cell-to-cell contact; T_C cells release perforins creating pores in target membranes; activates macrophages.
Target Pathogens	Extracellular bacteria, viruses in circulation, circulating venoms, toxins.	**Intracellular pathogens** (*Mycobacterium tuberculosis*, viruses inside cells, fungi, parasites) and cancerous cells.
Organ Graft Rejection	Plays a minor secondary role (hyperacute humoral rejection).	**Primary Mediator of Graft Rejection!** Allograft rejection is mediated by T_C cells recognizing non-self MHC/HLA antigens.
Immunosuppression	Intravenous immunoglobulins (IVIg), plasmapheresis.	**Cyclosporin A** (fungal metabolite from *Trichoderma polysporum*) inhibits calcineurin, suppressing T-cell activation.

Antibody Architecture (H_2L_2) & Immunoglobulin Classes

Each antibody is a Y-shaped heterodimer composed of **4 polypeptide chains**: two identical Heavy (H) chains (≈ 50 kDa each) and two identical Light (L) chains (≈ 25 kDa each), cross-linked by **interchain disulfide bonds** ($-S-S-$).

$$\text{Antibody Formula: } H_2L_2 \quad (\text{Two Heavy Chains H} + \text{Two Light Chains L}) \tag{1}$$

Fab (Antigen-Binding Fragment) contains Variable (V_H, V_L) domain; Fc mediates effector binding.

Class	Valency / Form	Diagnostic Role & High-Yield NCERT Facts
IgG	Monomer (Bivalent)	Most abundant ($75-80\%$ of serum Igs); **only antibody class that crosses human placenta** providing natural passive immunity to fetus.
IgA	Dimer (Tetravalent)	Secretory Ig linked by J-chain and secretory piece. Abundant in **Colostrum** (first yellowish milk), saliva, tears, respiratory mucous.
IgM	Pentamer (Decavalent)	Largest molecular weight, linked by J-chain; first antibody synthesized during primary immune response; potent complement activator.
IgE	Monomer (Bivalent)	Mediates **Type I Hypersensitivity (Allergies)**; binds Fc receptors on mast cells and basophils, triggering release of histamine and serotonin.
IgD	Monomer (Bivalent)	Co-expressed with IgM on the surface of naive mature B-cells; functions as antigen receptor initiating B-cell differentiation.

Lymphoid Organs: Primary vs Secondary Tier Architecture

- **Primary Lymphoid Organs (Generative/Central):** Sites where immature lymphocytes differentiate and proliferate into antigen-sensitive lymphocytes.
  - **Bone Marrow:** Primary site of hematopoiesis; produces all blood cells including B- and T-progenitors. B-cells mature completely here.
  - **Thymus:** Lobed organ situated near heart beneath breastbone; large at birth, undergoes progressive atrophy with age. Provides microenvironment for **T-lymphocyte maturation and thymic selection**.
- **Secondary Lymphoid Organs (Peripheral):** Sites where lymphocytes interact with antigens and proliferate into effector/memory cells.
  - **Spleen:** Large bean-shaped organ containing lymphocytes and phagocytes. Acts as a physiological blood filter trapping blood-borne microbes; serves as a massive **reservoir of erythrocytes (RBCs)**.
  - **Lymph Nodes:** Small solid structures along lymphatic vessels; filter lymph fluid by trapping microorganisms, presenting them to resident dendritic cells and lymphocytes.
  - **MALT (Mucosa-Associated Lymphoid Tissue):** Located within mucosal lining of major tracts (digestive, respiratory, urogenital). Constitutes $\approx 50\%$ **of all lymphoid tissue in the human body!** Includes Peyer's patches (ileum) and palatine tonsils.

2 Major Human Pathogens, Transmission & Diagnostic Tests

Bacterial, Viral, Protozoan & Helminthic Infectious Diseases

Disease	Causative Organism	Transmission Vector	Diagnostic Pathology & NCERT Key Hallmarks
Typhoid	*Salmonella typhi* (Gram-negative bacillus)	Contaminated food and water; classic carrier: **Mary Mallon** (Typhoid Mary).	Sustained high fever ($39-40^{\circ}$C), weakness, stomach pain, constipation, headache, anorexia. **Intestinal perforation** in severe cases. Confirmatory test: **Widal Test**.
Pneumonia	*Streptococcus pneumoniae*, *Haemophilus influenzae*	Inhaling droplets/aerosols; sharing glasses/utensils.	**Infects pulmonary alveoli**; alveoli become fluid-filled leading to severe respiratory distress. Lips and fingernails turn **gray to bluish** in severe cases.
Common Cold	**Rhino viruses** (ss-RNA)	Inhaling droplets; contaminated pens, books, cups.	**Infects nose and respiratory passage, but NOT lungs!** Nasal congestion, discharge, sore throat, hoarseness, cough; lasts $3-7$ days.
Amoebiasis	*Entamoeba histolytica* (Protozoan parasite)	**Houseflies** act as mechanical carriers from feces to food.	Infects large intestine; constipation, abdominal pain, cramps, stools with **excess mucous and blood clots**.
Ascariasis	*Ascaris lumbricoides* (Intestinal roundworm)	Ingestion of water, vegetables, fruits contaminated with fecal eggs.	Internal bleeding, muscular pain, fever, anemia, and mechanical **blockage of the intestinal passage**.
Filariasis (Elephantiasis)	*Wuchereria bancrofti*, *W. malayi* (Filarial worms)	Bite of infected **female *Culex*** mosquito.	Chronic inflammation of organs, especially **lymphatic vessels of lower limbs and genital organs**, resulting in gross gross deformities.
Ringworm	*Microsporum*, *Trichophyton*, *Epidermophyton*	Direct contact, sharing towels, clothes, comb; heat and moisture favor growth.	Dry, scaly, pruritic (itchy) circular lesions on skin, nails, scalp, and groin folds.

Life Cycle of *Plasmodium* (Malarial Parasite: Digenetic Cycle)

Plasmodium requires two hosts: **Primary host** = Female *Anopheles* mosquito (sexual cycle); **Secondary host** = Human (asexual cycle).

1. **Inoculation:** Female *Anopheles* mosquito injects infective stage **Sporozoites** with saliva into human blood during feeding.
2. **Hepatic Schizogony:** Sporozoites invade liver parenchymal cells, multiply asexually, and burst hepatocytes, releasing merozoites.
3. **Erythrocytic Schizogony:** Merozoites penetrate RBCs, form ring-stage trophozoites $\rightarrow$ schizonts, and burst RBCs synchronously every $48-72$ hours.
   - Rupture of RBCs releases pyrogenic toxin **Haemozoin** (formed from digested hemoglobin iron-porphyrin), producing **chills and high recurring fever**.
4. **Gametocytogenesis:** Some merozoites in RBCs develop into sexual stages: male microgametocytes and female macrogametocytes.
5. **Mosquito Phase (Sexual Cycle):** Female *Anopheles* ingests gametocytes with blood meal. Fertilization and zygote formation occur in the **mosquito gut**. Zygote becomes motile **ookinete**, penetrates gut wall to form **oocyst**. Oocyst divides by sporogony, generating thousands of sporozoites which migrate to and lodge in the **mosquito salivary glands**.

HIV/AIDS Pathogenesis & Viral Replication Mechanism

Caused by **Human Immunodeficiency Virus (HIV)**, an enveloped Lentivirus (Retroviridae) harboring **two identical single-stranded RNA (+ssRNA)** molecules and **Reverse Transcriptase**.

1. **Host Cell Attachment:** Viral envelope glycoprotein gp120 binds to CD4 **receptors** and chemokine coreceptors (CCR5/CXCR4) on macrophages and Helper T (T_H) lymphocytes.
2. **Macrophage Phase ("HIV Factory"):** Viral RNA is converted into viral DNA by Reverse Transcriptase (RNA $\rightarrow$ cDNA $\rightarrow$ dsDNA). Viral DNA integrates into host genome via **Integrase**. Macrophages produce viral particles without immediate cell lysis.
3. T_H **Cell Depletion:** Virus enters $CD4^+T_H$ cells, replicates, and lyses host cells. Helper T-cell count plummets from normal ($900-1200/\mu L$) to below **200**/μL.
4. **Opportunistic Infections:** Severe cell-mediated immunodeficiency permits life-threatening infections by *Mycobacterium tuberculosis*, *Toxoplasma gondii*, *Pneumocystis jirovecii*, and cytomegalovirus (CMV).
5. **Diagnosis & Management:** Screening test: **ELISA** (Enzyme-Linked Immunosorbent Assay); Confirmatory test: **Western Blot** (detects viral antigens p24, gp120, gp41). Treatment: Anti-Retroviral Therapy (**ART**) cocktail of nucleoside reverse transcriptase inhibitors (NRTIs, e.g., Zidovudine/AZT) and protease inhibitors.

Cancer Biology: Oncogenesis, Characteristics & Clinical Therapies

- **Transformation Hallmarks:** Normal cells exhibit **contact inhibition** (contact with neighboring cells inhibits uncontrolled proliferation). Cancer cells lose contact inhibition, proliferating relentlessly into **tumors**:
  - **Benign Tumors:** Encapsulated, remain localized to primary site, do not invade adjacent tissues.
  - **Malignant Tumors:** Unencapsulated masses of neoplastic cells exhibiting **Metastasis** (cells slough off into circulation and colonize distant organs; **the most feared property of malignant neoplasms**).
- **Etiological Carcinogens:**
  - **Physical:** Ionizing radiation (X-rays, γ-rays break DNA strands); Non-ionizing radiation (**UV-B** induces cyclobutane pyrimidine/thymine dimers).
  - **Chemical:** Tobacco smoke contains ***N*-nitrosamines** and benzo[*a*]pyrene (major cause of bronchogenic lung carcinoma).
  - **Biological:** Oncogenic viruses carrying viral oncogenes (*v-onc*); cellular proto-oncogenes (*c-onc*) mutated into active oncogenes.
- **Therapeutic Modalities:** Surgery (tumor excision); Radiotherapy (targeted lethal irradiation of malignant tissue); Chemotherapy (antimitotic drugs like Vincristine, Vinblastine from *Catharanthus roseus*); **Immunotherapy:** Biological response modifiers like **α-Interferon** activate the patient's immune system to destroy tumors.

Psychoactive Drugs, Hallucinogens & Addiction Pharmacology

Class	Representative Drugs	Receptors / Site	Pharmacological Action & Botanical Extraction
Opioids	Morphine, Heroin (**Smack** / Diacetylmorphine), Codeine	Specific opioid receptors in **CNS and GIT**.	Extracted from latex of unripe capsules of ***Papaver somniferum*** (Opium poppy). Heroin is synthesized by **acetylation of morphine**; white, odorless, bitter crystalline compound. Acts as a **depressant and slows down body functions**.
Cannabinoids	Marijuana, Hashish, Charas, Ganja	Cannabinoid receptors primarily located in the **brain**.	Obtained from inflorescences, leaves, and resin of ***Cannabis sativa*** (Hemp plant). Inhaled or orally ingested; induces relaxation, alters sensory perception, and severely impacts **cardiovascular system**.
Coca Alkaloids	**Cocaine** (Coke / Crack)	Interferes with reuptake transport of **Dopamine**.	Extracted from leaves of South American shrub ***Erythroxylum coca***. Potent CNS stimulant producing euphoria and increased energy; high doses induce **hallucinations**. Other hallucinogens: *Atropa belladonna*, *Datura*.
Tobacco / Nicotine	Nicotine (Alkaloid)	Stimulates nicotinic acetylcholine receptors.	Stimulates **adrenal medulla** to release adrenaline and noradrenaline into circulation, raising blood pressure and heart rate. Induces lung cancer, emphysema, bronchitis; elevates carbon monoxide (CO) in blood, decreasing oxyhemoglobin concentration.

3 Evolutionary Biology: Origins, Evidence & Mechanisms

Origin of Life: Abiogenesis & Miller-Urey Experimental Proof

- **Primordial Earth Atmosphere:** Formed ≈ 4.5 billion years ago. Primitive atmosphere was strictly **reducing**: no free O_2, rich in CH_4, NH_3, H_2O (vapor), H_2.
- **Chemical Evolution (Oparin-Haldane Hypothesis):** First forms of life originated from pre-existing non-living organic molecules (RNA, proteins, sugars), preceding biological evolution.
- **Miller-Urey Spark-Discharge Experiment (1953):** Stanley Miller simulated prebiotic atmospheric conditions:
  - Closed spark-discharge flask containing gases: $CH_4 : NH_3 : H_2$ in ratio $\mathbf{2} : \mathbf{1} : \mathbf{2}$ along with continuously boiling water vapor (H_2O).
  - Tungsten electrodes produced continuous electrical discharge ($\approx 75,000\,\text{V}$) maintaining chamber temperature at **800**°C.
  - **Experimental Yield:** Condensed effluent contained amino acids: **Glycine, α-Alanine, and Aspartic acid**, conclusively validating chemical evolution.

Comparative Morphological Evidences: Homology vs Analogy

Parameter	Homologous Organs (Divergent Evolution)	Analogous Organs (Convergent Evolution)
Fundamental Basis	**Same anatomical origin and basic structure**, modified to perform different functions in response to different adaptations.	**Different anatomical origin and basic structure**, modified to perform similar functions due to similar selective pressures.
Evolutionary Pattern	**Divergent Evolution** (Radiates from a common ancestor into diverse ecological niches).	**Convergent Evolution** (Unrelated lineages evolve functionally convergent phenotypes).
Vertebrate Forelimb Examples	Forelimb skeletal architecture of **Whale (flipper), Bat (patagium), Cheetah (running), and Human (grasping)** shares identical arrangement: humerus, radius, ulna, carpals, metacarpals, phalanges.	**Wings of Butterfly** (chitinous membranous folds) vs **Wings of Birds** (feather-covered forelimb bones).
Sensory / Functional Organs	Vertebrate heart and brain anatomy across fish, amphibians, reptiles, birds, and mammals.	**Eye of Octopus** (invertebrate retina derived from skin) vs **Eye of Mammals** (derived from embryonic diencephalon). **Flippers of Penguins (bird)** vs **Dolphins (mammal)**.
Botanical Examples	**Thorns of *Bougainvillea*** (axillary bud protection) vs **Tendrils of *Cucurbita*** (axillary bud climbing).	**Sweet Potato** (root tuber modification) vs **Potato** (underground stem tuber modification).

Adaptive Radiation & Convergent Marsupial-Placental Counterparts

Adaptive Radiation: Evolution of divergent species starting from a common ancestral point and radiating into distinct geographical habitats and ecological niches.

- **Classic Examples:** Darwin's Finches on Galapagos archipelago (ancestral seed-eating beak diversified into vegetarian, insectivorous, warbler, and cactus-puncturing beaks); Australian Marsupials radiating from a single ancestral stock within the isolated island continent.

- **Marsupial vs Placental Mammal Convergence:** When adaptive radiation occurs in isolated continents for different lineages occupying similar ecological niches, convergent evolution generates striking morphological analogs:

Ecological Niche	Australian Marsupial Counterpart	Placental Mammal Counterpart
Fossorial Burrower	Marsupial mole (*Notoryctes*)	Placental Mole
Myrmecophage (Ant-eater)	Numbat / Banded anteater (*Myrmecobius*)	Anteater
Small Glider	Flying phalanger / Sugar glider (*Petaurus*)	Flying squirrel
Feline Carnivore	Tasmanian tiger cat (*Dasyurus*)	Bobcat
Apex Carnivore	Tasmanian wolf (*Thylacinus*)	Wolf (*Canis lupus*)

Hardy-Weinberg Equilibrium: Mathematical Formulations & Selection Dynamics

In a large, randomly mating, panmictic diploid population with no evolutionary disruptive forces, allele frequencies and genotype frequencies remain constant across generations:

$$\mathbf{p + q = 1} \qquad \Longrightarrow \qquad \mathbf{p^2 + 2pq + q^2 = 1} \tag{2}$$

Where p = frequency of dominant allele A, q = frequency of recessive allele a, p^2 = frequency of homozygous dominant AA, $2pq$ = heterozygous Aa, and q^2 = homozygous recessive aa.

- **Five Evolutionary Forces Disrupting Equilibrium:** (1) Gene migration / Gene flow; (2) **Genetic Drift** (random allele frequency fluctuation in small populations, including **Founder Effect** and **Bottleneck Effect**); (3) Mutation; (4) Genetic Recombination during meiosis; (5) Natural Selection.
- **Three Modes of Natural Selection on Phenotypic Traits:**
  1. **Stabilizing Selection:** Favors intermediate phenotype ($2pq$); phenotypic curve becomes narrower and taller (e.g., human infant birth weight centered at $3 - 3.5$ kg).
  2. **Directional Selection:** Favors one extreme phenotype; shifts population mean towards one direction (e.g., Industrial Melanism in *Biston betularia*, antibiotic resistance in bacteria).
  3. **Disruptive Selection:** Favors both extreme phenotypes over intermediate; produces a bimodal, two-peaked curve.

Chronological Geological Timeline of Human Evolution

Ancestor Stage	Geological Era	Fossil Provenance	Cranial Capacity	Anatomical & Cultural Characteristics
Dryopithecus	$\approx$ 15 mya (Miocene)	Africa and Eurasia	$\approx$ 350 – 400 cc	Arboreal, hairy, arms longer than legs; walked quadrupedally like modern gorillas and chimpanzees.
Ramapithecus	$\approx$ 15 mya (Miocene)	Shivalik hills (India)	$\approx$ 400 – 450 cc	More man-like than *Dryopithecus*; upright posture tendency; walked semi-bipedally.
Australopithecus	$\approx 2 - 3$ mya (Pliocene)	East African grasslands	$\approx$ 450 – 600 cc	**Bipedal locomotion**, upright posture; hunted with stone weapons but **essentially ate fruit**.
Homo habilis	$\approx 1.7 - 2$ mya (Pleistocene)	Olduvai Gorge (Tanzania)	**650 – 800** cc	**First human-like hominid** ("Tool maker" / Handy man); constructed crude stone tools; **did not eat meat**.
Homo erectus	$\approx$ 1.5 mya (Pleistocene)	Java (1891 fossils)	**900** cc	Fully erect posture; proficient toolmaker; first to master **fire**; **ate meat**.
Homo neanderthalensis	100,000 – 40,000 ya	Neander Valley (Germany)	**1400** cc	Heavy brow ridges, receded chin; **used animal hides to protect bodies; buried their dead with rituals**.
Homo sapiens	75,000 – 10,000 ya	Emerged in Africa	**1350 – 1450** cc	Modern man; migrated across continents; prehistoric cave art ($\approx$ 18,000 ya at Bhimbetka); **agriculture began** $\approx$ 10,000 **ya**.

High-Frequency NEET Assertion-Reason Traps in Health & Evolution

- **Rhino Virus vs Streptococcus:** Rhino viruses infect the **upper respiratory tract and nasal passage, strictly sparing the pulmonary alveoli**; *Streptococcus pneumoniae* infects the alveoli causing fluid accumulation and cyanosis.
- **Malarial Host Nomenclature:** Sexual reproduction occurs **only in the female *Anopheles* mosquito** (the definitive/primary host). Asexual multiple fission occurs in humans (intermediate host).
- **Morphine vs Heroin:** Morphine is a natural alkaloid; Heroin is a **semi-synthetic derivative produced by acetylation of morphine**.
- **Darwin vs de Vries on Variation:** Darwinian variations are **small, continuous, and directional**; de Vriesian mutations are **single-step large mutations (Saltation), random, and directionless**.
- **Meat Consumption in Hominids:** *Homo habilis* **did not eat meat**; *Homo erectus* **ate meat**!